BRIEFING TO THE DEAN · FKTEN

Design Project Digitalisation at FKTEN

Progress so far, and the way forward to a faculty platform for project-based and industry-based courses

NMK31704 DESIGN PROJECT TEAM

Ts. Dr. Mohd Zamri Zahir Ahmad

Ts. Dr. Ahmad Ashraf Abdul Halim · Ts. Dr. Aini Syuhada Md Zain · Dr. Nurol Husna Che Rose

PROGRAMME REPRESENTATIVES

Engineering (NMJ32004) **Diploma** (NDJ30304) **BTech** (WBL & InTra): Dr. Iszaidy Bin Ismail

DATE

20 October 2026

FACULTY

Faculty of Electronic Engineering & Technology

Universiti Malaysia Perlis

TODAY

What we would like to share

1. **Where we started**
   The course, its scale, and two semesters of manual administration

2. **What we achieved** — LIVE DEMO
   Guidebook, three systems, results and user feedback, Gold Award

3. **Next semester: one unified DPMS** — LIVE DEMO
   Ready for Semester 2, 2026/2027 on the current platform

4. **The way forward**
   One faculty platform for Engineering, Diploma and BTech, on the university's server

5. **Resources and decisions**
   Team, 18-month timeline, budget and what we ask today

PART 1

Where we started

NMK31704 Design Project, Bachelor of Electronic Engineering Technology

THE COURSE

NMK31704 Design Project

- **Group-based design project** offered every Semester 2
- **100% continuous assessment**, no final examination
- **4 to 5 students** in a group, mixed across programmes
- **1 or 2 supervisors** for each group, plus **2 panels** for the final assessment
- **14 weeks**, run by **4 course coordinators**

THREE ENGINEERING TECHNOLOGY PROGRAMMES, ONE COURSE

UR6523007	Bachelor of Electronic Engineering Technology (Electronic Systems) with Honours
UR6523008	Bachelor of Electronic Engineering Technology (Electronic Telecommunication Design) with Honours
UR6523009	Bachelor of Electronic Engineering Technology (Electronic Network Design) with Honours

Students, supervisors and panels all come from these three programmes, so every decision has to be balanced across them.

THE SCALE

One semester, in numbers

182
students

43
project groups

80+
lecturers as supervisors and panels

592
weekly logbooks to verify and mark

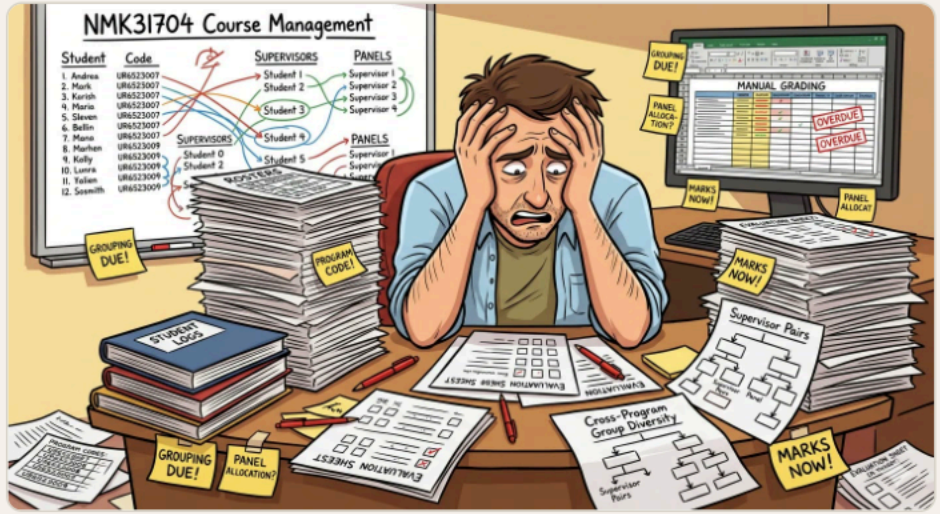
258
evaluations to collect from supervisors and panels

12
rubrics feeding 7 course outcomes

Semester 2, 2025/2026. Evaluations counted as six per group: report by the supervisor and two panels, presentation by two panels, supervisor observation.

THE PROBLEM

For two semesters, all of it was manual



- ✕ **Grouping:** student records sorted by hand, then checked again for every rule
- ✕ **Logbook:** written in MS Word, saved as PDF, signed by the supervisor, submitted in eLearning
- ✕ **Peer review:** Google Form, then manual queries in Google Sheets to filter and sort each student's scores
- ✕ **Assessment:** a Google Sheet made to look like a system. One sheet per group, so supervisor and panels could see each other's marks, and every panel change was edited by hand
- ✕ **Final materials:** each group sent its report, poster and video to its panels by email and eLearning

Semester 2, 2023/2024
Manual

Semester 2, 2024/2025
Manual

Semester 2, 2025/2026
Automated ecosystem

PART 2

What we achieved

Guide

one clear handbook for students, supervisors and panels

Automate

grouping, supervisor and panel allocation

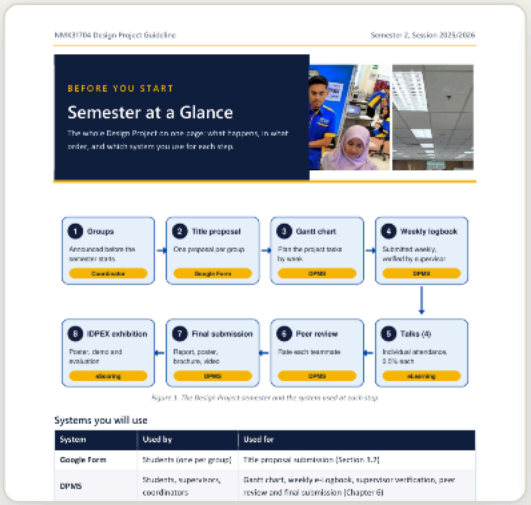
Centralise

logbooks, submissions and assessment online

A course handbook rewritten around the systems



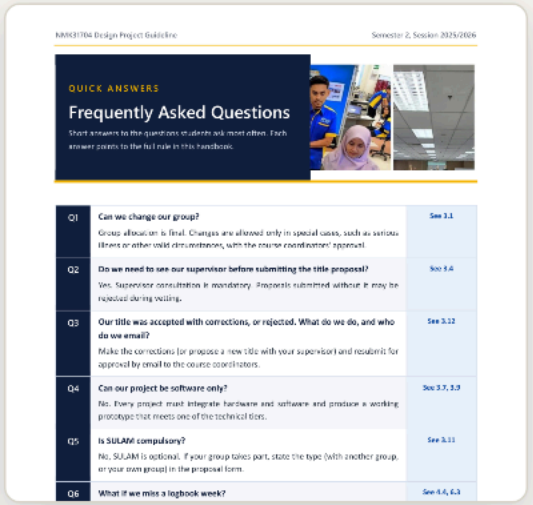
- ✓ **Semester at a glance:** every step, who acts, and which system is used
- ✓ **Who evaluates what:** each component, its marks and its evaluator in one table
- ✓ **Quick answers:** the questions students ask most, each pointing to the rule
- ✓ **Step-by-step appendices** for the e-logbook, the Gantt chart and every rubric



Semester at a glance



Who evaluates what



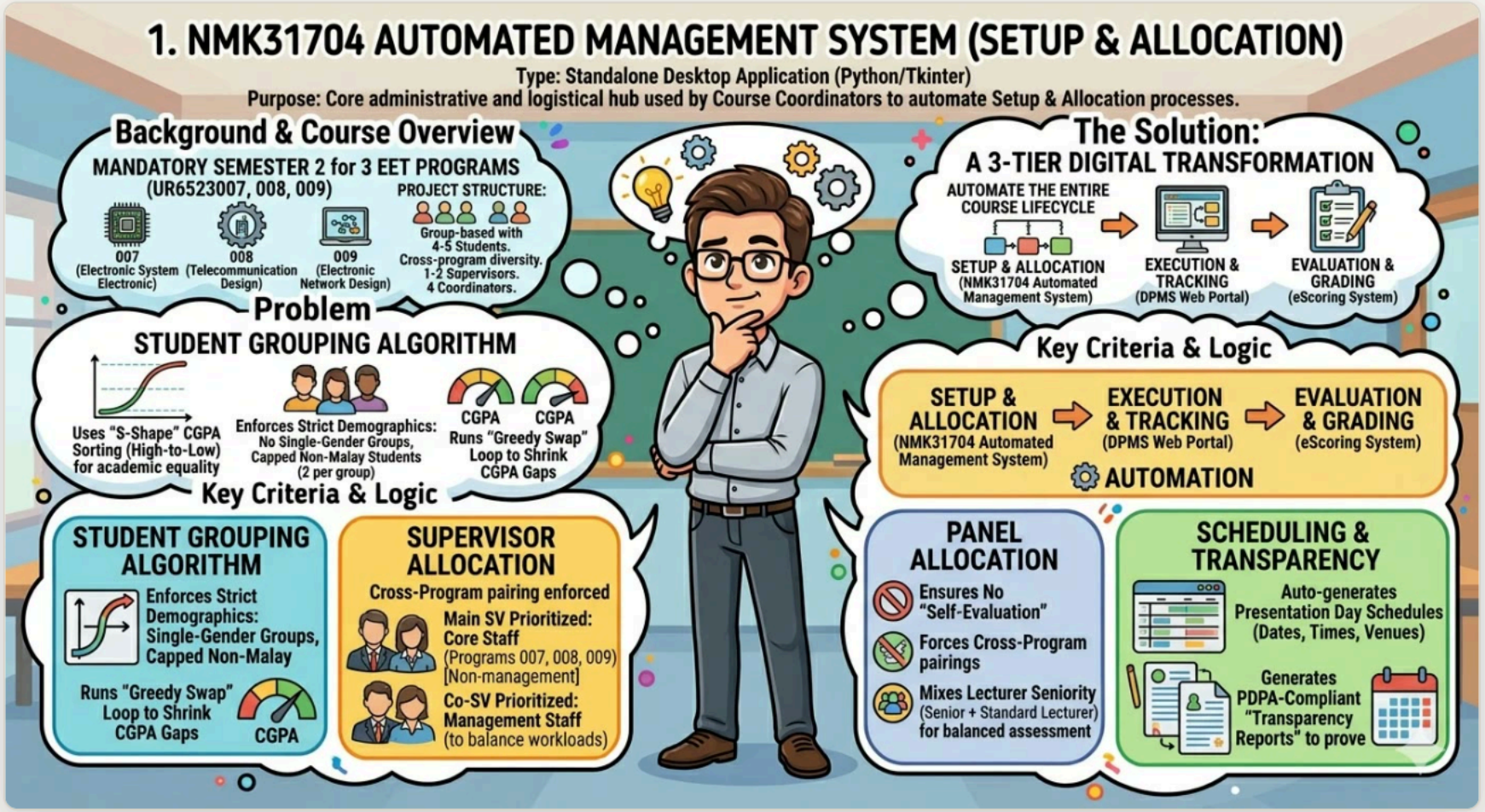
Quick answers

One ecosystem, three systems, at a glance

1 · SET UP
Automated Management System
Desktop application

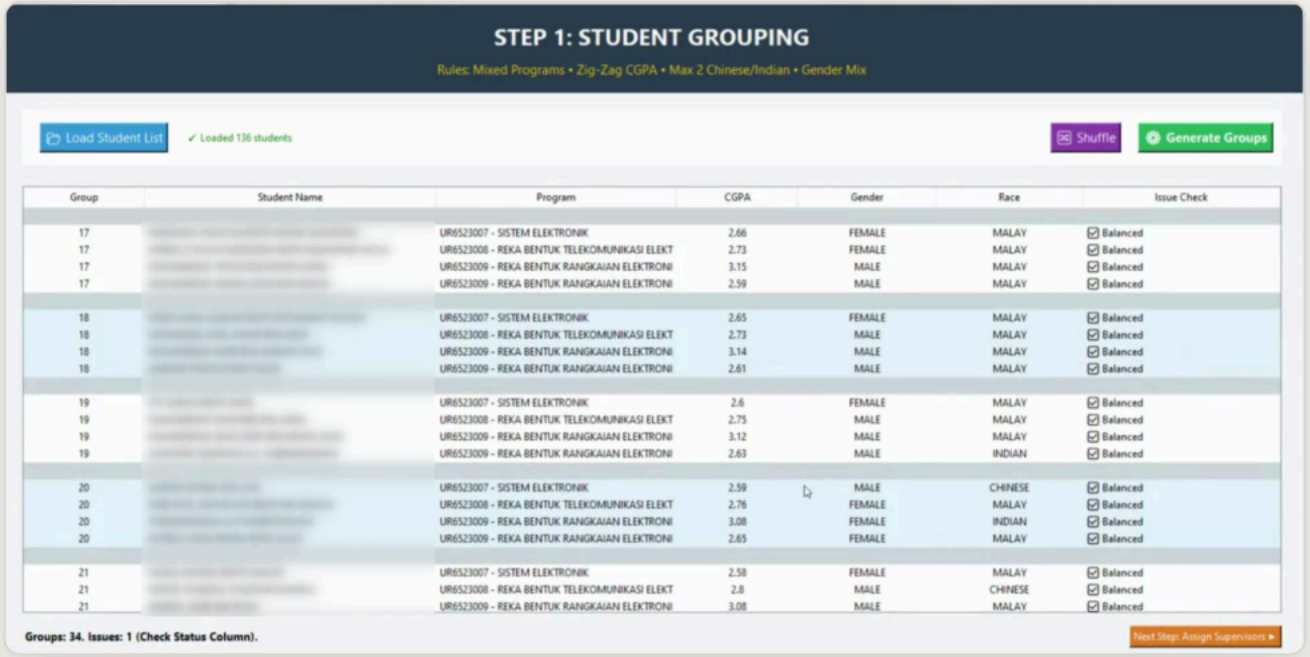
2 · RUN
Design Project Management System (DPMS)
Web application

3 · ASSESS
eScoring System
Web application

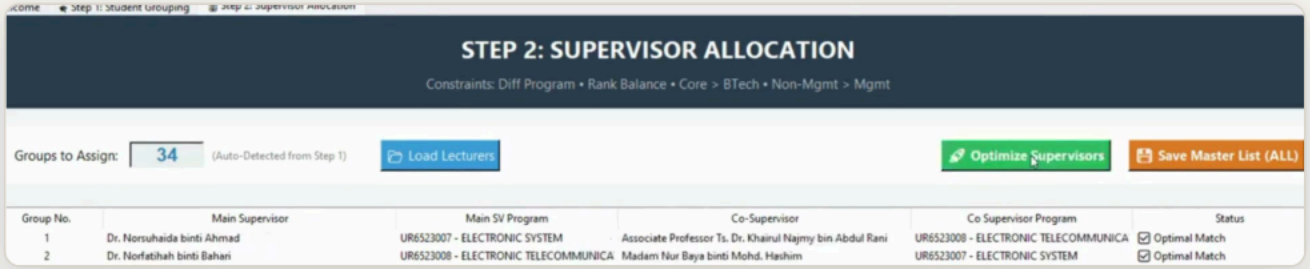


SYSTEM 1 · SET UP

Balanced groups and fair allocation in minutes



Step 1: load the student list, generate groups, every row checked. Click to enlarge.



Step 2: supervisor pairs from different programmes

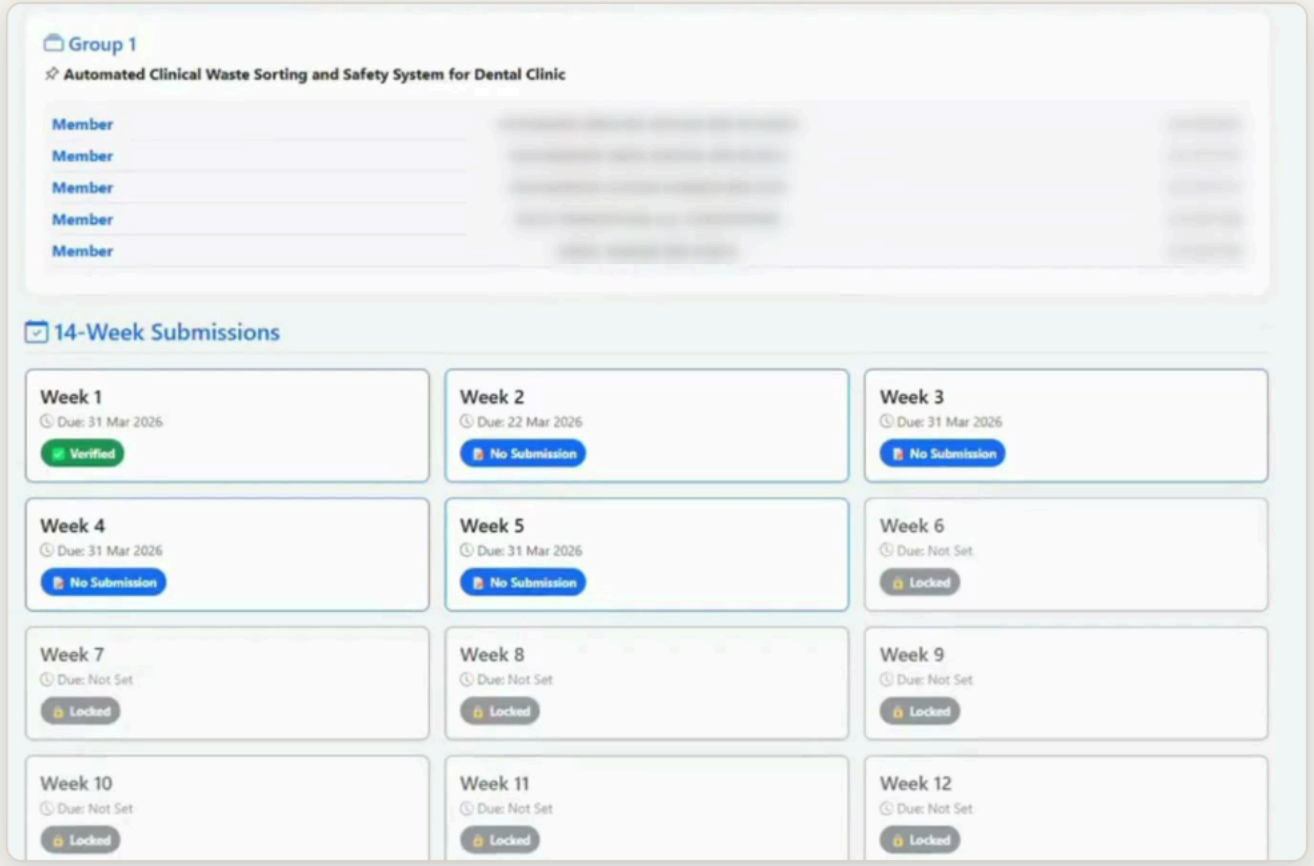
RULES THE SYSTEM ENFORCES

- ✓ Every group mixes the three programmes
- ✓ CGPA, gender and race balanced across groups
- ✓ Supervisor and co-supervisor from different programmes
- ✓ Management staff assigned as co-supervisor
- ✓ No panel evaluates a group they supervise

HOW IT WORKS

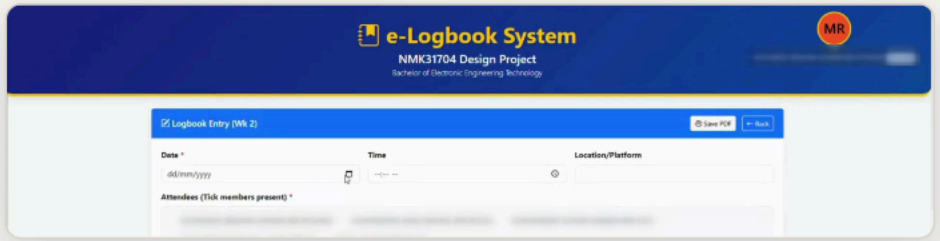
Students are ordered by CGPA and dealt into groups in a zig-zag. The system then swaps students between groups until the gaps are small, and reports how the allocation was made.

DPMS: one place for the whole 14 weeks



A student's page: the group, and the status of each week's logbook. Click to enlarge.

- ✓ **Weekly logbook** filled in on the web, with photos as evidence
- ✓ **Gantt chart** planned once, then ticked through the logbook each week
- ✓ **Supervisor verification** on screen, with email notification
- ✓ **Peer review** confidential, with marks calculated by the system
- ✓ **Final submission** of report, poster, brochure and video link
- ✓ **Coordinators mark logbooks** inside the same system



The weekly logbook form

eScoring: mark on the spot, track it live

EVALUATOR
My assigned groups
Role, venue, session, countdown

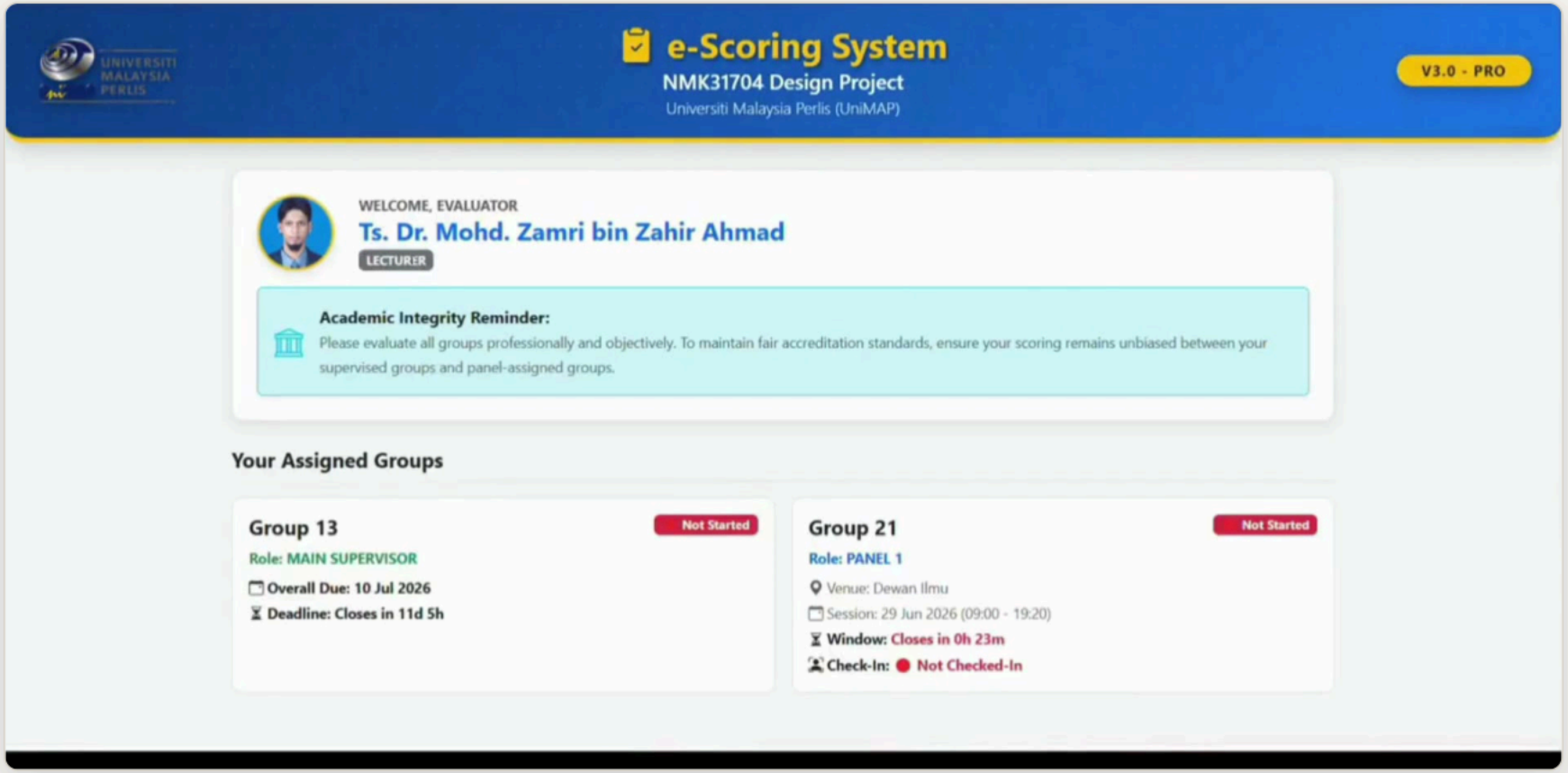
EVALUATOR
Check-in at the venue
Unlocks presentation marking

EVALUATOR
Marking with the rubric
Guide and live mark per criterion

EVALUATOR
Completed
Status and live marks summary

COORDINATOR
Evaluation status
Supervisor and both panels, per group

COORDINATOR
Live check-in monitor
Who has arrived, by venue



THE IMPACT

Before and after

TASK	MANUAL, TWO SEMESTERS	WITH THE ECOSYSTEM
Form 43 balanced groups, assign supervisors and panels	Sorted by hand in a spreadsheet **about 2 to 3 working days**	Generated and checked by rules **minutes**
Collect, verify and mark 592 logbooks	Word, PDF, signature, eLearning, download one by one **about 30 hours of handling**	Submitted, verified and marked on one page **no file handling**
Peer review marks	Google Form, then filter and sort about 600 responses student by student	Rated in the system, marks calculated **instant**
Final materials to the panels	About 170 files sent by email and eLearning	Panels open them from the group page
Marks, grades, CO/PO and audit evidence	One shared sheet per group, queries rebuilt by hand **about 1 to 2 working days**	Private to each evaluator, calculated on submission **instant**
Who has not submitted or marked	Ask, wait, ask again	Live status for every group and evaluator

Counts are from Semester 2, 2025/2026. Time figures and the number of peer review responses are estimates by the coordinators, not measured values.

THE IMPACT

What 41 users told us

EASE OF USE FOR THEIR TASKS

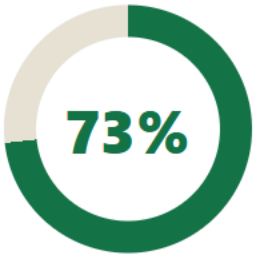
8.5 / 10

33 of 41 gave 8 or higher

STREAMLINES MANAGEMENT AND ASSESSMENT

8.6 / 10

35 of 41 gave 8 or higher



NO ISSUES WHILE USING IT

30 of 41. The other 11 reported an issue, mostly signing in and phone use.

MOST USEFUL FEATURE

Online logbook named by 18 of 31 who answered, then the **Gantt chart** and **supervisor verification**.

WHAT THEY ASKED FOR NEXT

Better phone support, more reliable sign-in, autosave for logbook drafts.

> *"Overall the system makes students work much easier."*

> *"Easy for students to update progress and supervisors can monitor project progress."*

> *"Good implementation to be used in future."*

Source: DPMS User Experience Survey on forms.app, 41 responses (34 students, 6 lecturers, 1 not stated), 29 May to 23 June 2026. share.forms.app/mohdzamrizahirahmad/design-project-management-system-dpms-dpms-user-experience-survey

RECOGNITION

Gold Award, EDIC 2026



Gold Award

Education Innovation Challenge (EDIC) 2026

International Conference on Quality, Sustainability & Innovation in Science and Technology Education (IQ-SITE 2026)

CATEGORY

Transformative Teaching

DATE

1 September 2026

TEAM

Mohd Zamri Zahir Ahmad, Ahmad Ashraf Abdul Halim, Nurol Husna Che Rose, Aini Syuhada Md Zain

LIVE DEMO

The system as used in Semester 2, 2025/2026

1. **Grouping tool:** generate balanced groups and supervisor pairs
2. **DPMS, student:** group page and weekly logbook
3. **DPMS, supervisor:** verify a logbook
4. **eScoring, panel:** mark with the rubric
5. **Coordinator:** live evaluation status and marks export

Backup if the network fails: the 6-minute video at youtu.be/Z26XAPRhVqI

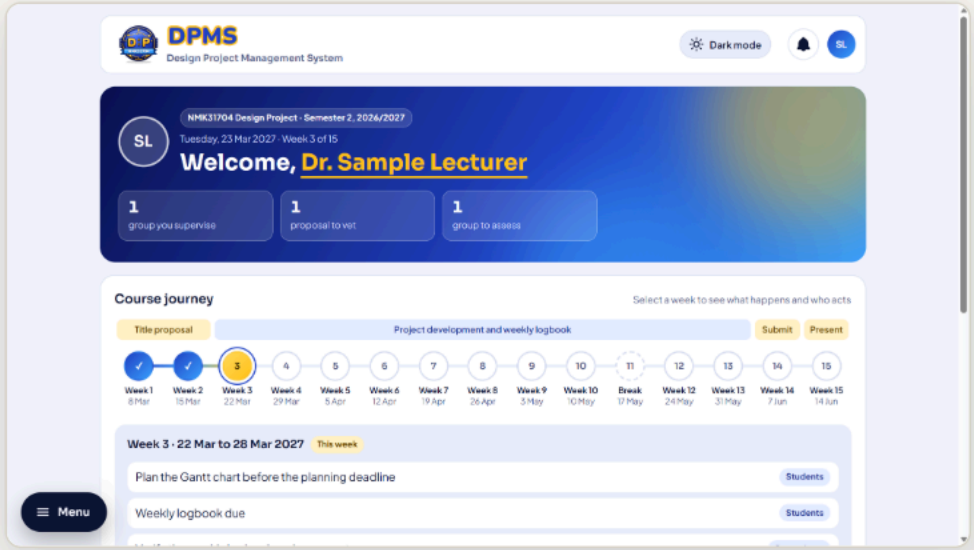
PART 3

Next semester

Three systems become one unified DPMS for Semester 2, 2026/2027

One unified DPMS, ready on the current platform

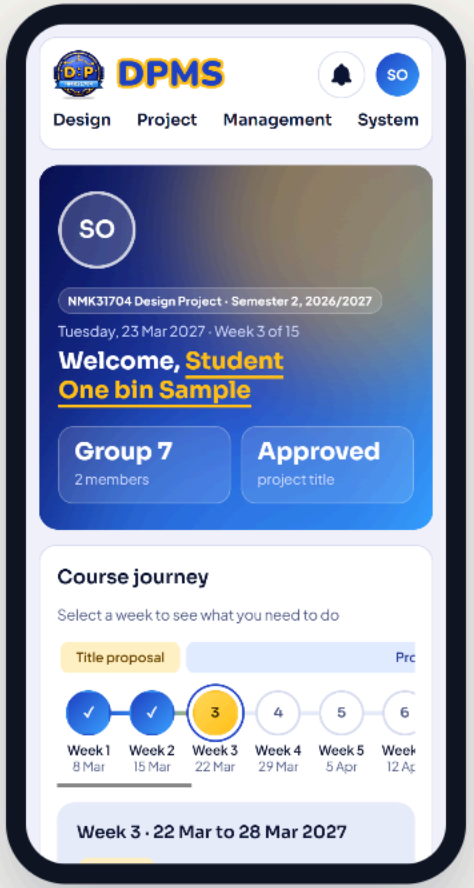
- ✓ **One system, one sign-in:** DPMS and eScoring merged; grouping results imported directly
- ✓ **New: title proposal and vetting,** blind review, accepted title goes straight to the group
- ✓ **Every role on one home page:** supervisor, vetter, panel and coordinator together
- ✓ **Works on a phone,** the most requested improvement in the survey
- ✓ **Demo copy and test group** for training without touching real data



A lecturer's home page: every role in one place (sample data)

1,098
automated checks pass on every release

60
database tables already designed for MySQL



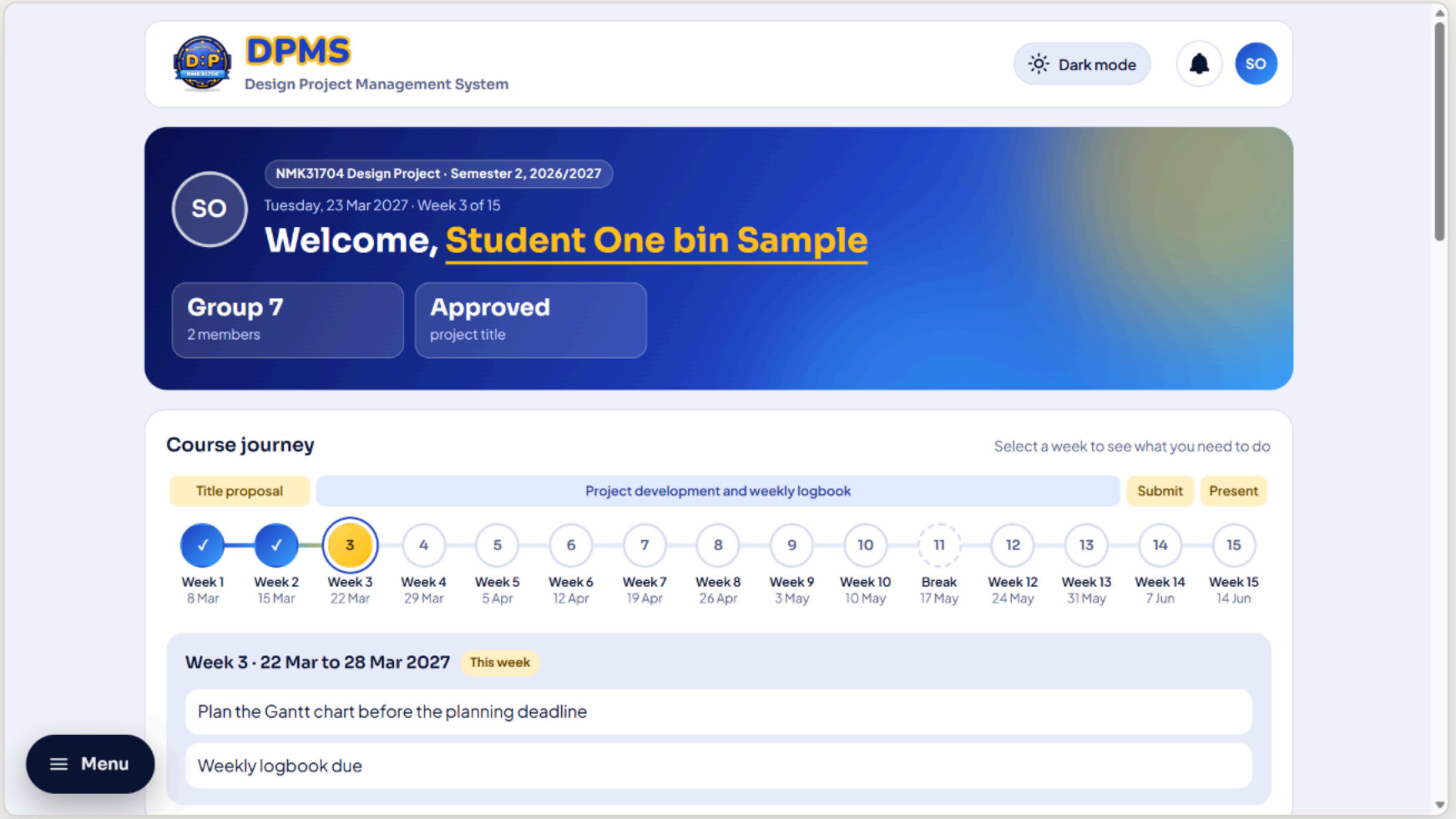
A look inside the unified DPMS

STUDENT
Home and course journey
Group, title, week by week

STUDENT
My logbook
14 weekly cards with status

VETTER
Blind title vetting
Sees the proposal, never the names

COORDINATOR
Semester set-up
Checklist, calendar, grouping import, assessment scheme



All screens use made-up sample people.

LIVE DEMO

The unified DPMS for Semester 2, 2026/2027

1. **Student:** home page, title proposal, weekly logbook
2. **Lecturer:** all roles on one home page; verify a logbook
3. **Vetter:** blind review of a title proposal
4. **Panel:** mark the report and presentation
5. **Coordinator:** semester set-up, monitoring and results
6. **Phone:** the same pages on a mobile screen

Shown with the built-in demo copy and sample data, so no real student record is used.

WHY THE NEXT STEP IS NEEDED

What the current platform cannot do

Only UniMAP Google accounts

UniMAP restricts Apps Script sharing, so industry panels, industry mentors and external examiners cannot sign in.

A spreadsheet as the database

Each read or write to Google Sheets takes seconds. Pages slow down when many panels mark at once on presentation day.

Built for one course

Rules, marks and rubrics follow NMK31704. Other courses assess differently and cannot use it as it is.

Not on university servers

Student records sit in Google Drive files instead of the university's own infrastructure managed by DMDC.

Other courses still manual

Engineering design project, Diploma FYP and BTech WBL still run on Word files, Google Forms and email.

Survey feedback

11 of 41 users reported an issue, mostly signing in and phone use. A proper web platform removes the sign-in limits.

PART 4

The way forward

One FKTEN platform for project-based and industry-based learning

Four programme groups, about 550 students a semester

CATEGORY	PROGRAMME	COURSE	STUDENTS / SEMESTER
Project-based	Bachelor of Electronic Engineering Technology (UR6523007, 008, 009)	NMK31704 Design Project	200
Project-based	Bachelor of Microelectronic, Biomedical Electronic and Electronic Engineering (UR6523001, 004, 005)	NMJ32004 Integrated Design Project	200
Project-based	Diploma in Electronic Engineering (UR4523003)	NDJ30304 Final Year Project	100
Industry-based	Bachelor of Technology in Industrial Electronic Automation (UR6523011)	NMT31304, NMT32205, NMT32310 (WBL) and Industrial Training (InTra)	50
Total			**550**

> Title: Development of an Integrated FKTEN Digital Platform for the Management and Assessment of Project-Based and Industry-Based Learning.

THE PROPOSAL

One server, two systems, four sites

Students | **Supervisors / lecturers** | **Staff panels** | **Industry panels & mentors** | **Coordinators / admin**

UniMAP sign-in for staff and students · email invitation or one-time code for industry

DMDC server · AlmaLinux · Apache · PHP 8.3 · MySQL/MariaDB · SSL

System A: project-based courses

One shared code base, configured per course

Engineering Technology	Engineering	Diploma
NMK31704 · /engtech	NMJ32004 · /eng	NDJ30304 · /diplomaelectronic
own database	own database	own database

System B: WBL and InTra

Its own code base

BTech
UR6523011 · /btech
own database

Proposed address: designprojectfkten.unimap.edu.my, subject to DMDC approval. Each site keeps its data separate; improvements to System A reach all three sites at once.

THE PROPOSAL

What each system will do

System A: project-based courses

1	Grouping and allocation balanced groups, supervisor and panel rules	PROVEN
2	Title proposal and vetting blind review, decision flows to the group	BUILT
3	Gantt chart and weekly logbook supervisor verification, coordinator marking	PROVEN
4	Peer review and final submission	PROVEN
5	Supervisor and panel assessment online rubrics, industry panel access	PROVEN
6	Results, iGrading and CO-PO attainment	PROVEN
7	Course set-up per programme components, weights, rubrics, CO-PO, panel style	NEW

System B: WBL and InTra (BTech)

1	Placement and eligibility companies, industry mentors, CGPA check	NEW
2	Student compliance documents undertaking, health declaration, insurance	NEW
3	Daily or weekly logbook verified by the industry mentor by link or code	NEW
4	Monitoring visits by lecturers	NEW
5	Report submission	NEW
6	Industry and university assessment cognitive, psychomotor and affective domains	NEW
7	Results, iGrading and CO-PO attainment	NEW

WBL and industrial training today

HOW AN ASSESSMENT FORM TRAVELS NOW

Course coordinator
prepares a Google Form for each assessment

→

Email
sends the form link to the WBL coordinator

→

WBL coordinator
compiles the forms and sends them to each company

→

Industry
fills in the forms; marks are compiled again by hand

WHO ASSESSES (WBL AND INTRA)



Most of the marks come from people who cannot sign in to a UniMAP-only system.

ALSO HANDLED SEPARATELY TODAY

- ✕ Undertaking, health declaration and insurance documents
- ✕ Logbooks and their verification by industry
- ✕ Mark compilation, 1 to 2 weeks

HOW IT WILL BE BUILT

From Apps Script to PHP and MySQL on the DMDC server

Now: Google Apps Script

- Unified DPMS runs NMK31704 in Semester 2, 2026/2027
- Google Sheets as the database
- UniMAP Google accounts only
- No cost, but limited speed and access

→

If approved: PHP + MySQL at DMDC

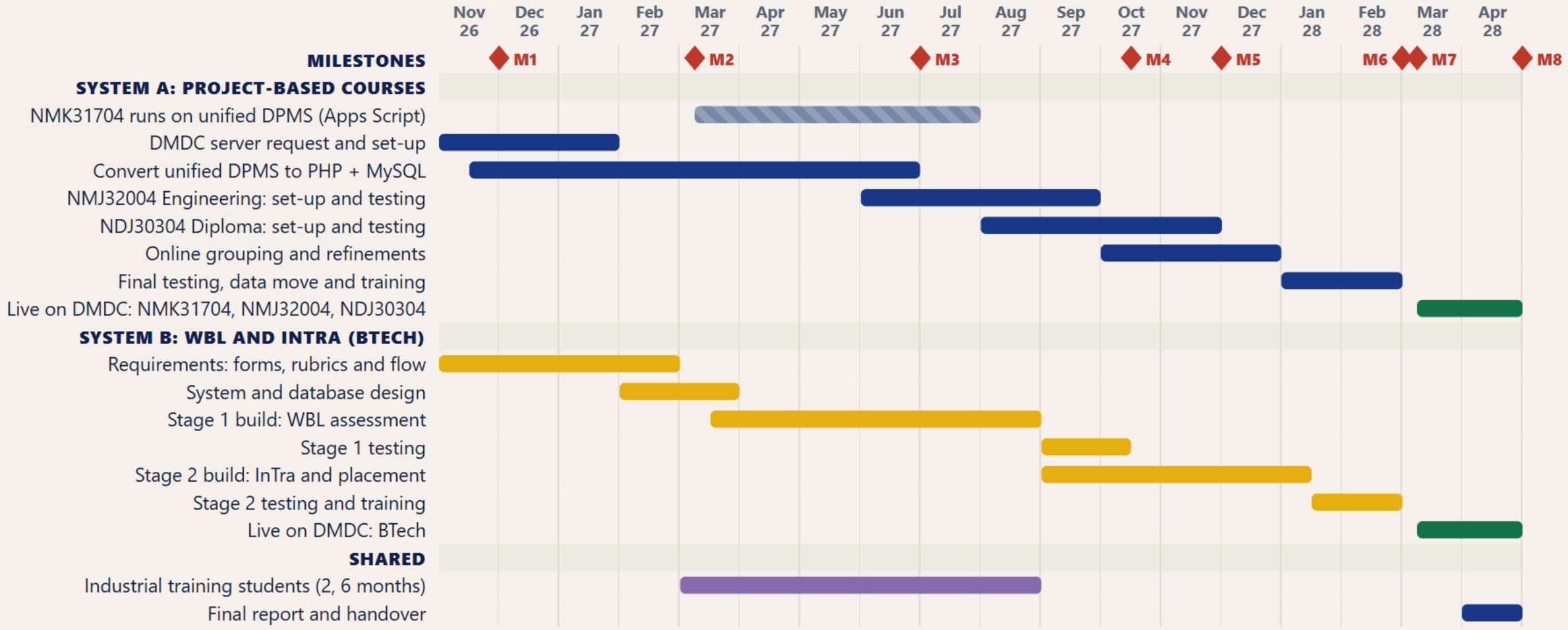
- The same workflows and screens, rewritten as PHP
- The 60-table design moved into MySQL
- The 1,098 checks reused to prove nothing changed
- Sign-in for industry; extended to three more programmes

WHY AN AI CODING SUBSCRIPTION IS NEEDED

The conversion and the new modules are done in-house by two lecturers with **Claude Code**, an AI coding assistant. It rewrites the Apps Script code as PHP, builds System B and runs the tests, while the developers design, review and decide. A monthly subscription gives a fixed, predictable cost; pay-per-use API billing cannot be budgeted. Development uses sample data only.

TIMELINE

18 months: November 2026 to April 2028



M1 Nov 26: design, DMDC request
M2 8 Mar 27: unified DPMS live
M3 Jul 27: PHP core on DMDC
M4 Oct 27: WBL tested
M5 Dec 27: Eng. and Diploma ready
M6 Feb 28: all tested, users trained
M7 Mar 28: all four sites live
M8 Apr 28: final report

TEAM

Built in-house by FKTEN lecturers

NAME	REPRESENTS	ROLE IN THE PROJECT
Ts. Dr. Mohd Zamri Zahir Ahmad	NMK31704 (Electronic Engineering Technology)	Project leader and developer of System A
Dr. Iszaidy Bin Ismail	WBL and InTra (Industrial Electronic Automation)	Developer of System B
Ts. Dr. Ahmad Ashraf Abdul Halim	NMK31704 (Electronic Engineering Technology)	Requirements and testing
Ts. Dr. Aini Syuhada Md Zain	NMK31704 (Electronic Engineering Technology)	Requirements and testing
Dr. Nurol Husna Che Rose	NMK31704 (Electronic Engineering Technology)	Requirements and testing
Programme representative	NMJ32004 (Microelectronic, Biomedical Electronic and Electronic Engineering)	Course coordinator, requirements and testing
Programme representative	NDJ30304 (Diploma in Electronic Engineering)	Course coordinator, requirements and testing
2 industrial training students	6 months	Testing, data entry and user support

RESOURCES

Budget requested: RM15,000

ITEM	QUANTITY	RM
AI CODING SUBSCRIPTION, ONE ACCOUNT PER DEVELOPER		
System A developer: Max 20x, conversion phase	3 months	2,660
System A developer: Max 5x	4 months	1,775
System A developer: Pro	11 months	975
System B developer: Max 5x, build phase	5 months	2,215
System B developer: Pro	9 months	800
OTHER		
Allowance, 2 industrial training students	RM500 × 6 months	6,000
Face-to-face training for coordinators	3 sessions	300
Contingency (exchange rate, price changes)	±2%	275
Total		**15,000**

WHAT IT BUYS

- Conversion of the unified DPMS to the university's server
- Two more project courses on the same code
- A new WBL and InTra system for BTech

WHAT IT DOES NOT NEED

- ✓ No vendor and no licence
- ✓ No new hardware: staff laptops
- ✓ Server from DMDC; backups in university Google Drive
- ✓ Staff training online

Subscription prices include 8% SST at RM4.10/USD (October 2026 prices).

RESOURCES

What we will request from DMDC

Server	Virtual machine or web hosting at the DMDC data centre
Software	AlmaLinux 9, Apache 2.4 with SSL, PHP 8.3, MySQL 8 or MariaDB, phpMyAdmin
Capacity	2 vCPU and 4 GB RAM minimum (4 vCPU and 8 GB recommended)
Storage	300 GB minimum (500 GB recommended) for reports, posters, documents and logbook photos
Address	designprojectfkten.unimap.edu.my with /engtech, /eng, /diplomaelectronic and /btech
Services	UniMAP sign-in (SSO) if available, SMTP relay for email, daily backup
Access	SSH/SFTP or Git for the developers

WHY THE UNIVERSITY'S SERVER

- Student records stay on university infrastructure (PDPA 2010)
- The same technology DMDC already runs for other faculty systems
- Faster than a spreadsheet, and open to industry users

TODAY

What we ask from the Dean

1. **Agreement in principle**
   so the working paper can go to the Faculty Management Committee

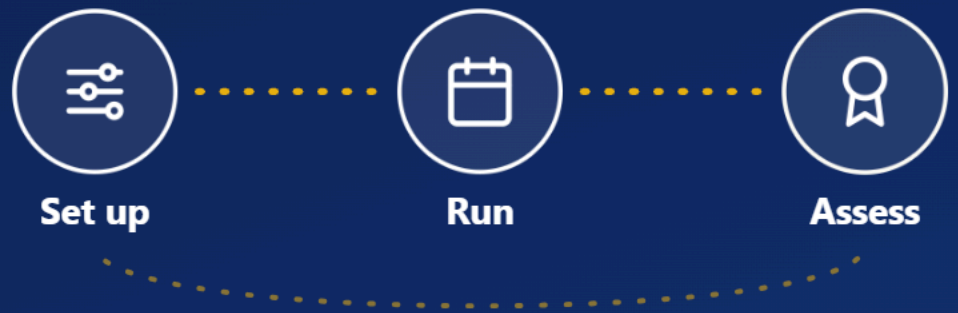
2. **Budget of RM15,000**
   and advice on the fund to use (proposed: FKTEN operating budget)

3. **Support for the DMDC request**
   server, subdomain and storage as listed

4. **The 18-month timeline**
   November 2026 to April 2028, with all four sites live in Semester 2, 2027/2028

Thank you

Questions and discussion



The DPMS site

THE TEAM

Ts. Dr. Mohd Zamri Zahir Ahmad · Ts. Dr. Ahmad Ashraf Abdul Halim
Ts. Dr. Aini Syuhada Md Zain · Dr. Nurol Husna Che Rose
with the programme representatives for Engineering, Diploma and BTech

SEE IT

sites.google.com/unimap.edu.my/dpfkten
youtu.be/Z26XAPRhVqI (6-minute video)